

EDUCATION

- IMDEA Software Institute, Universidad Politécnica de Madrid (UPM)

10/2022 - Present

PhD Student

Madrid, Spain

– Software, Systems and Computing. Supervisor: Thaleia Dimitra Doudali.

– Thesis Title: Smart and Sustainable Cloud Resource Management with Minimal Use of Machine Intelligence.

– Expected Graduation Date: October, 2027.
- National Technical University of Athens (NTUA)

09/2017 - 09/2022

Bachelor’s and Integrated Master’s Degree - Electrical and Computer Engineering (ECE)

Athens, Greece

– GPA: 8.38/10. 300 ECTS.

– Specialization: Computer Systems, Software, Biomedical Engineering.

PUBLICATIONS

- Aceso: Latency-Aware Geo-Distribution of Microservices for Cost-Effective Carbon Reduction

Georgia Christofidi, Francisco Alvarez Terribas, Ioannis Roumpos, Jesus Alberto Omana Iglesias, Nicolas Kourtellis, Thaleia Dimitra Doudali

– To appear in the 27th ACM International Middleware Conference (Middleware 2026).

– <https://arxiv.org/pdf/2603.10768>
- Advancing Workload Management with Foundational Models: Challenges in Time Series Similarity and Interpretability.

Tiemo Bang, Ioannis Roumpos, Sergiy Matuskevych, Georgia Christofidi, Yuanyuan Tian, and Thaleia Dimitra Doudali.

– In Proceedings of the 1st workshop connecting academia and industry on Modern Integrated Database and AI Systems (MIDAS ’25). Colocated with SIGMOD’25. Association for Computing Machinery, New York, NY, USA, 16–25.

– <https://doi.org/10.1145/3737412.3743491>
- Augmenting Cloud Resource Management with the Necessary Amount of Machine Intelligence

Georgia Christofidi, Thaleia Dimitra Doudali.

– Euro-Par 2024. Lecture Notes in Computer Science, vol 15386. Springer, Cham.

– https://doi.org/10.1007/978-3-031-90203-1_38
- Do Predictors for Resource Overcommitment Even Predict?

Georgia Christofidi and Thaleia Dimitra Doudali.

– In Proceedings of the 4th Workshop on Machine Learning and Systems (EuroMLSys ’24). Colocated with EuroSys’24. Association for Computing Machinery, New York, NY, USA, 153–160.

– <https://doi.org/10.1145/3642970.3655838>
- Is Machine Learning Necessary for Cloud Resource Usage Forecasting?

Georgia Christofidi, Konstantinos Papaioannou, and Thaleia Dimitra Doudali.

– In Proceedings of the 2023 ACM Symposium on Cloud Computing (SoCC ’23). Association for Computing Machinery, New York, NY, USA, 544–554.

– <https://doi.org/10.1145/3620678.3624790>
- Toward Pattern-based Model Selection for Cloud Resource Forecasting.

Georgia Christofidi, Konstantinos Papaioannou, and Thaleia Dimitra Doudali.

– In Proceedings of the 3rd Workshop on Machine Learning and Systems (EuroMLSys ’23). Colocated with EuroSys’23. Association for Computing Machinery, New York, NY, USA, 115–122.

– <https://doi.org/10.1145/3578356.3592588>

PROFESSIONAL EXPERIENCE

- Google

06/2025 - 09/2025

Full-time Student Researcher in Systems Research@Google (SRG)

Seattle, WA, USA

– Improving Lifetime-Aware VM scheduling.

– Hosts: Yawen Wang, Martin Maas.
- Telefónica Digital Innovation

06/2024 - 12/2024

Full-time Research Intern

Barcelona, Spain

– Distributed Systems, Edge, Carbon Efficiency.
- National Observatory of Athens

01/2022 - 09/2022

Master’s Thesis and Full-time Internship

Penteli, Greece

– Distributed Systems, Edge, Carbon Efficiency.

– Under the supervision of the European Space Agency (ESA).

PATENTS

•Method, Orchestrator and Computer Programs for Workload Management Across Cloud and Edge Computing Environments

Georgia Christofidi, Jesus Alberto Omana Iglesias, Nicolas Kourtellis, Francisco Alvarez Terribas

- Applicant/Owner: Telefónica Innovación Digital S.L. European Patent Office (EPO), Application No. EP25382227.
- Date: 12 March 2025. Status: filed patent application.

SCHOLARSHIPS - AWARDS

- **Awards:** Best Talk Award at the Machine Learning for Edge-Cloud Systems (ML4ECS) workshop, colocated with HiPEAC 2025. 3rd place at the 2026 Annual Poster Competition at the IMDEA Software Institute.
- **PhD Grant:** A. G. Leventis Foundation
- **Travel Grant Awards:** SOSP’26, EuroPar’24, womENCourage’24, SoCC’23.

SUMMER SCHOOLS - TRAINING

- Very Large Databases (VLDB) Summer School** 07/2024
Cluj - Napoca, Rumania
- Artificial Intelligence Summer School** 07/2023
Center for Artificial Intelligence and Machine Learning (CAIML) - 2 ECTS Points Vienna, Austria
- Deep Learning and Computer Vision Summer School** 06/2023
Machine Learning Genoa Center (MaLGa) Genoa, Italia

TALKS - PRESENTATIONS

- 32nd Symposium on Operating Systems Principles (SOSP’26).** 10/2026
Poster Presentation - Carbon-Aware and Cost-Effective Geo-Distribution for Latency-Critical Microservices. Prague, Czech Republic
- 20th European Conference on Computer Systems (EuroSys’25).** 04/2025
Poster and Doctoral Workshop - Maximizing Return On Investment for Sustainable Operations. Rotterdam, Netherlands
- Machine Learning for Edge-Cloud Systems (ML4ECS) workshop. Colocated with HiPEAC 2025.** 01/2025
Oral Presentation - CaRE: Towards Carbon and Resource Efficient Orchestration at the Cloud-Edge Continuum. Barcelona, Spain
- womENCourage™ 2024 - ACM Celebration of Women in Computing:** 06/2024
Poster Presentation. University Carlos III of Madrid (UC3M). Madrid, Spain
- GEC ’23 – 5th Summit on Gender Equality in Computing** 06/2023
Poster Presentation. Athens University of Economics and Bussiness (AUEB). Athens, Greece
- Feminine Innovation: Women and Girls in Montegancedo event** 02/2023 and 2024
AI Workshop for primary school students. Universidad Politécnica de Madrid (UPM). Madrid, Spain
- Fireballs Workshop 03 on Fireball Databases and Machine Learning:** 08/2022
Open-Source Software presentation. Flash Detection Software - Neliota Project. Glasgow, UK

SERVICE

- **Artifact Evaluation Committee Member:**
 - 18th and 19th USENIX Symposium on Operating Systems Design and Implementation (OSDI 2024 and 2025)
 - 2024 USENIX Annual Technical Conference (2024 USENIX ATC)
 - 29th Architectural Support for Programming Languages and Operating Systems (ASPLOS 2024)
 - 19th and 18th European Conference on Computer Systems (EuroSys’24 and 23 Fall AE)
 - 6th Conference on Machine Learning and Systems (MLSys 2023)
- **Shadow PC Member:** 20th European Conference on Computer Systems (EuroSys’25)
- **Student Representative at IMDEA Software Institute:** 2024-2025
- **Volunteer at Madrid is Science Event:** 2023, 2024